

Stellar

Colour changes in mid-B star variables

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Abstract: Model predictions of light changes due to non-radial stellar pulsation have been compared with the observations of light and colour changes in mid-B star variables made by Waelkens and Rufener (1985). It is concluded that these observations are consistent with the presence of low l (dipole and quadrupole) non-radial pulsations in the stars. This strengthens Waelkens and Rufener's hypothesis that these variables may be identified with the 53 Persei stars. The differences between the model predictions for these large Q variables, which have periods in the 1 to 3 day range, and the predictions of small Q models, which apply to β Cephei variables, are explained.

Introduction

The 53 Per variables are a loose grouping of early-B star variables identified by Smith (1980a) as non-radial g-mode pulsators. This identification was based on observed periods and line profile variations which suggested the presence of low l oscillations. More recently, Waelkens and Rufener (1985) have observed light and colour changes in seven mid-B star variables of spectral types in the range B4 to B8 and with luminosity classes III and IV. They determine periods in the one to three day range. Because both the spectral type and periods partly overlap those of objects studied by Smith and his co-workers, Waelkens and Rufener suggest that these mid-B star variables be identified with the 53 Per class. If this identification is correct, then the Geneva $\Delta[U-B]$ data which they present are the first substantial sample of colour change data for 53 Per variables that includes ultraviolet observations. This is important in that U-band data are of particular value in hot variables when colour changes are used to examine the question of possible non-radial pulsation.

Waelkens and Rufener, however, note a possible problem with their observed $\Delta[U-B]$ data. This is that the observed colour changes exceed those expected from low l non-radial pulsation if the early work of Stamford and Watson (1977) on β Cep variables is used as a guide. Here this problem is examined. Non-radial models are used to predict light and colour changes applicable to the mid-B star variables observed by Waelkens and Rufener. Because the Q values of the mid-B star variables exceed those of β Cep stars by more than an order of magnitude, it is shown that the predictions of the non-radial models for the two cases exhibit some differences. It is then

demonstrated that there is no conflict between the $\Delta[U-B]$ data and the hypothesis of non-radial pulsation in the mid-B star variables.

The Non-Radial Model

Light changes due to small non-radial pulsations may be expressed in an analytical form. This was initially demonstrated by the work of Dziembowski (1977), which was later extended by Buta and Smith (1979), Balona and Stobie (1979) and Stamford and Watson (1981). Here we follow the model presented in the Stamford and Watson paper, with some modifications and notational changes. In this form, the observed light change for a mode of index (l, m) is, in magnitudes,

$$\Delta m(\lambda, t) = -1.086 \epsilon P_{l, |m|}(\mu_0) [(T_1 + T_2) \cos(\omega t + \psi_T) + (T_3 + T_4 + T_5) \cos(\omega t)]$$

where

$$T_1 = b_{l\lambda} \cdot \alpha_T(\lambda) \cdot B$$

$$T_2 = \frac{\partial b_{l\lambda}}{\partial \log T_e} \cdot \frac{B}{2.3026}$$

$$T_3 = b_{l\lambda} (2+l)(1-l)$$

$$T_4 = -b_{l\lambda} \cdot p^* C \cdot \alpha_g(\lambda)$$

$$T_5 = -\frac{\partial b_{l\lambda}}{\partial \log g} \cdot \frac{p^* C}{2.3026}$$

$$\text{and where } b_{l\lambda} = \int_0^1 h \cdot \mu \cdot P_l(\mu) d\mu$$

$$\alpha_T(\lambda) = \frac{\partial \log F_\lambda}{\partial \log T_e}$$

$$\alpha_g(\lambda) = \frac{\partial \log F_\lambda}{\partial \log g}$$

$$p^* \approx \left(\frac{\partial \log g}{\partial \log p_g} \right)_{\tau=1}$$

$$C = (4 + \alpha_H^{-1}) - l(l+1)\alpha_H$$

$$\alpha_H = G \rho_0 Q^2 / 3\pi$$

$$B = R B_{\text{adiab}} = R(1 - \Gamma_2^{-1})C$$

In this representation of $\Delta m(\lambda, t)$, the terms which are frequently of most importance are T_1 and T_3 , which account for local temperature and geometry changes. The T_4 term attempts to account for local pressure changes. The terms T_2 and T_5 permit the inclusion of the variation of limb-darkening properties with changes in local $(T_e, \log g)$ over the stellar

surface. T_2 and T_3 are generally not significant if $l \leq 2$, which is the limit to the discussion in this paper. It is hoped to present elsewhere a detailed discussion of the $\Delta m(\lambda, t)$ model and its applicability to various stellar groups, so the following remarks will be confined to those aspects which are necessary for the present problem only.

Six of the parameters appearing in $\Delta m(\lambda, t)$ are fixed by published stellar atmosphere data at the $(T_e, \log g)$ of the star or group of stars in question. These are: $\alpha_T(\lambda)$ and $\alpha_g(\lambda)$, which give the local flux dependence on T_e and $\log g$ respectively; $b_l \lambda$, a weighted integral of the normalized limb-darkening law involving Legendre polynomials; the temperature and gravity derivatives of b_l ; and p^* , a measure of the variation of atmospheric gas pressure with surface gravity in layers giving rise to the observed continuum flux. The Kurucz (1979) atmospheres and the related Wade & Rucinski (1985) limb-darkening data were the source of these six parameters for the present work.

The amplitude parameter ϵ , the m index of the associated Legendre polynomial $P_l, |m|$, and μ_0 , the cosine of the inclination of the stellar rotation axis to the observer, are all unknown. These three unknowns can, however, be avoided when comparing the model with data in an attempt to determine l . This is achieved by writing $\Delta m(\lambda, t) = A_\lambda \cos(\omega t + \phi_\lambda)$ for each of two observed wavelengths or filter bands and then making the comparison with the data on an amplitude ratio versus phase difference plane, $(A_{\lambda_2}/A_{\lambda_1}, \phi_{\lambda_2} - \phi_{\lambda_1})$. Alternatively, a colour amplitude and phase may be derived from the two filter bands so that model and data may be compared on a $(A_{col}/A_{\lambda_1}, \phi_{col} - \phi_{\lambda_1})$ plane. Here the latter plane is employed.

If the Q value of the pulsation is estimated from a knowledge of the oscillation period of the variable, then R and ψ_T are the only two parameters that remain to be assigned before a model value of $(A_{col}/A_{\lambda_1}, \phi_{col} - \phi_{\lambda_1})$ is determined for each oscillation l value. These parameters R and ψ_T measure the amplitude and phase of local effective temperature changes with respect to local radius changes. In an adiabatic situation $(R, \psi_T) = (1, \pi)$. Non-adiabatic effects mean that (R, ψ_T) values may be poorly known, so here an outlined 'area of interest' on the $(A_{col}/A_{\lambda_1}, \phi_{col} - \phi_{\lambda_1})$ plane is presented for each l . This was done by assigning two possible values to each of R and ψ_T as an indication of the effect of uncertainty in these parameters. The R values of 1.0 and 0.25 were chosen as representative ones, following the discussion of Stamford and Watson (1981) in relation to different types of radially pulsating variables. The ψ_T values of $\pi \pm \pi/4$ were chosen, based on a knowledge that the models of Davey (1970) and Saio & Cox (1980) predict ψ_T values close to π for the β Cep variables both in radial and low l non-radial pulsation. The larger period mid-B star variables might, on the basis of the discussion of 53 Per stars by Smith (1980a), be expected to exhibit more extreme non-adiabatic conditions than the β Cep variables.

One final point in connection with the $\Delta m(\lambda, t)$ model concerns the expression for C . This C value, which relates local gas pressure changes to local radius changes, is always positive for radial pulsation but may be either positive or negative in non-radial pulsation. In the positive case, the usual radial

pulsation situation applies because vertical motions dominate and the non-radial pulsation. Regions of local contraction have highest pressures and, in the adiabatic situation, also highest effective temperatures. In the negative case, locally contracted regions have lowest pressures because horizontal motions in the non-radial pulsation have dominated and material has flowed away from these regions.

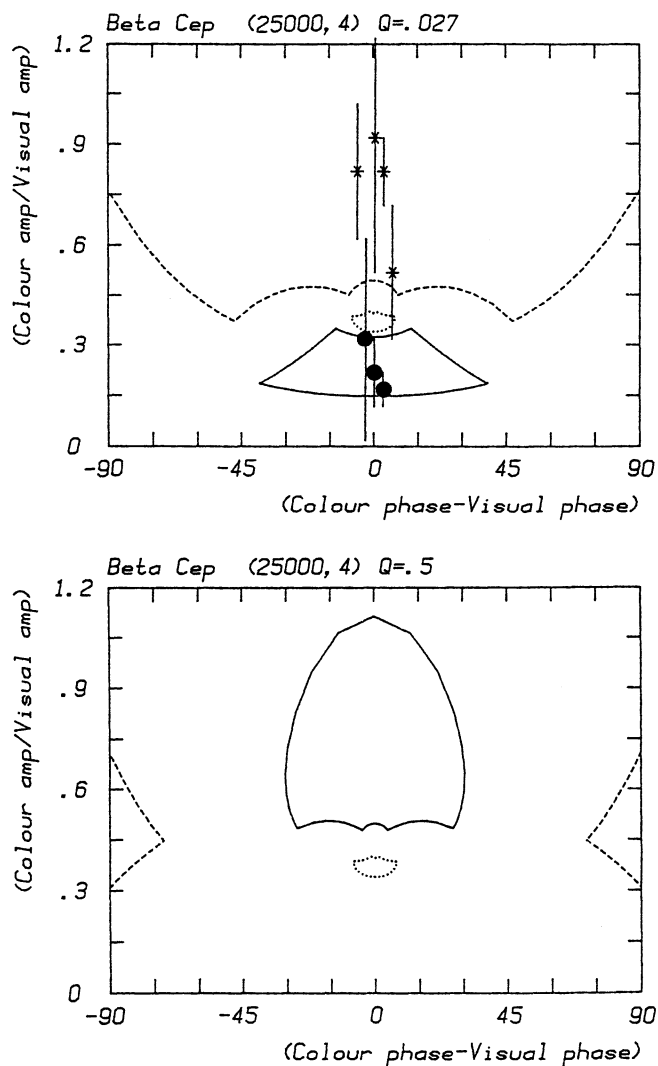


Figure 1 — A_{U-V}/A_V versus $(\phi_{U-V} - \phi_V)$ for non-radial pulsation models at $(T_e, \log g) = (25\,000, 4)$, typical of a β Cep variable. The upper panel is for $Q = 0.027$, characteristic of first overtone radial β Cep pulsation (Shobbrook 1985). The lower panel shows the effect of increasing Q to 0.5, typical of the mid-B star variables. Areas of interest, based on R values of 1 to 0.25 and ψ_T values of $5\pi/4$ to $3\pi/4$ are outlined with dashed curves for $l=0$, dotted curves for $l=1$ and full curves for $l=2$. On this basis, permitted regions lie inside the closed curves in the $l=1$ and 2 cases and, for the $l=0$ example, above the curve in the upper panel or outside the small sectors in the lower panel. Data are plotted for some β Cep systems which have been assessed independently from line profile studies as radial pulsators (asterisks) or non-radial pulsators (filled circles).

Comparison with Data

It has been known since the early work of Stamford and Watson (1977) that A_{U-V}/A_V values for quadrupole modes in β Cep variables permit these modes to be distinguished from radial modes. Thus, before turning to the question of the mid-B star variables, a β Cep model together with observations is given in Figure 1, in order to confirm the format used here for the presentation of the $\Delta m(\lambda, t)$ calculations. In this figure, curves based on R values of 1 and 0.25 and ψ_T values of $\pi \pm \pi/4$ indicate areas of interest for $l=0, 1$ and 2 in a model whose stellar atmosphere data apply to $(T_e, \log g) = (25\,000, 4)$. In the upper panel of Figure 1, a value of $Q=0.027$, which is characteristic of first overtone radial pulsation of β Cep variables, was used. If (R, ψ_T) values fall within the limits adopted, then only the region above the dashed line in the upper panel of Figure 1 is permitted for radial models. The corresponding permitted regions for $l=1$ and $l=2$ are within the dotted closed curve and the full closed curve, respectively. This suggests that if $(\phi_{U-V} - \phi_V)$ is close to zero, which is what is observed in β Cep stars, then $A_{U-V}/A_V > 0.45$ will favour radial pulsations and $A_{U-V}/A_V < 0.35$ will favour quadrupole pulsations. Dipole pulsations occupy a small intermediate region.

Independent mode assignments for β Cep variables are available through the line profile work of Campos and Smith (1980) and Smith (1980b, 1983). For those modes for which A_{U-V} data are also available in the literature, points have been plotted in Figure 1. Nominal values of $(\phi_{U-V} - \phi_V) = 0$ were employed since specific phase data are generally not available, though light and colour variations are referred to as being 'in phase'. All cases where an $l=0$ assignment was made from the line profiles fall in the $l=0$ region here. All three cases where a non-radial assignment was made from the line profiles fall in the $l=2$ region of Figure 1. Two of these had specific l values assigned in the line profile studies and in both these cases, which are modes in ν Eri and 12 Lac, this assignment was $l=2$. Thus, the format used here for the presentation of the $\Delta m(\lambda, t)$ model is fully consistent with the line profile studies and earlier claims on the usefulness of A_{U-V}/A_V data for mode discrimination in β Cep variables.

The lower panel of Figure 1 shows the effect of increasing Q to 0.5, a value which is close to that expected for the Waelkens and Rufener (1985) mid-B star variables, given their observed periods and spectral classes. Using the same (R, ψ_T) values as in the upper panel of Figure 1, it is seen that, whereas the dipole region of interest is not altered, the radial and quadrupole regions change substantially. Specifically, the $l=2$ region has moved from A_{U-V}/A_V values smaller than in the dipole region to values larger than in the dipole region. Similarly, the permitted lower limit to A_{U-V}/A_V values for $l=0$, along the $(\phi_{U-V} - \phi_V) = 0$ line, has receded up off the graph to > 2.0 .

Figures 2 and 3 present the mid-B star data of Waelkens and Rufener in a similar manner to Figure 1, using the same (R, ψ_T) values to indicate areas of interest for each l . In both cases, the lower panels use the $Q=0.5$ of the mid-B star variables, and the upper panels demonstrate the effect of decreasing Q by an order of magnitude. Stellar atmosphere data at $(T_e, \log$

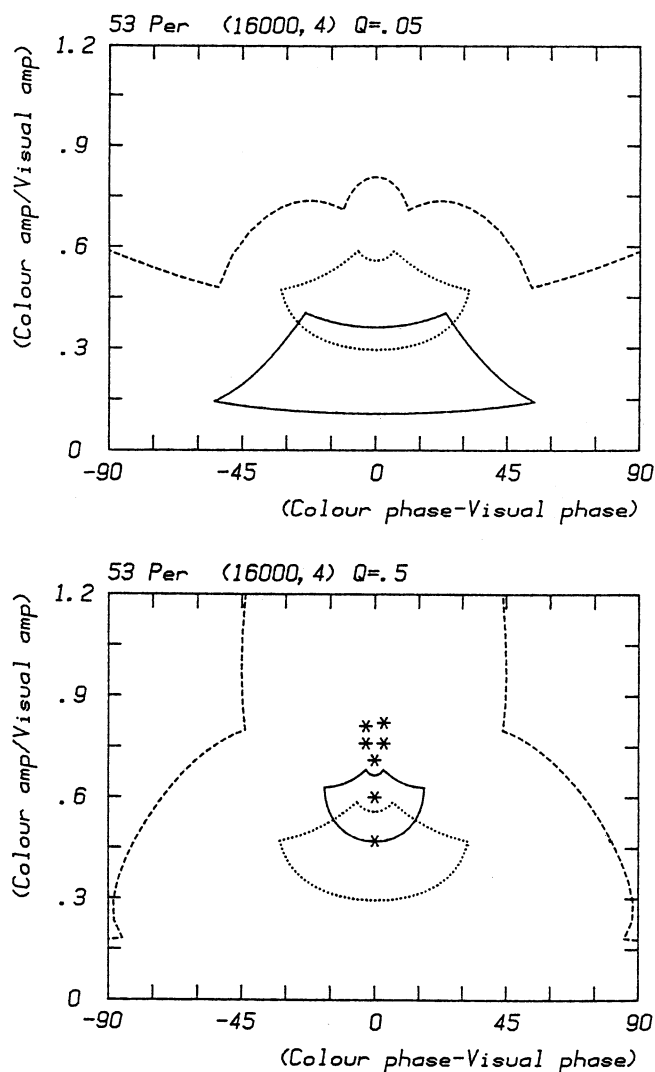


Figure 2— A_{U-V}/A_V versus $(\phi_{U-V} - \phi_V)$ for non-radial pulsation models at $(T_e, \log g) = (16\,000, 4)$, typical of a cooler 53 Per variable. The lower panel is for $Q=0.5$, characteristic of the mid-B star variables. The upper panel simply demonstrates the effect of decreasing Q by an order of magnitude. Areas of interest for $l=0, 1$ and 2 are outlined as explained in Figure 1. Data for mid-B star variables, derived from Waelkens and Rufener (1985) observations, are plotted.

$g) = (16\,000, 4)$ was used. This is characteristic of a cool 53 Per variable or the hotter end of the stars examined by Waelkens and Rufener. While details may alter, the general form of the changes of areas of interest with changing Q is not sensitive to the T_e of the model selected. The comments already made in connection with the $(25\,000, 4)$ model apply equally to this $(16\,000, 4)$ model and to even cooler models.

The separate presentations in Figures 2 and 3 reflect two different attempts to deal with less than ideal assumptions that must be made by virtue of the fact that the Waelkens and Rufener photometry is on the Geneva system, and is $\Delta[U-B]$ data. In Figure 2, as was the case in Figure 1, the model $\alpha_T(\lambda)$ and $\alpha_g(\lambda)$ data are based on Kurucz (1979) Johnson filter data to

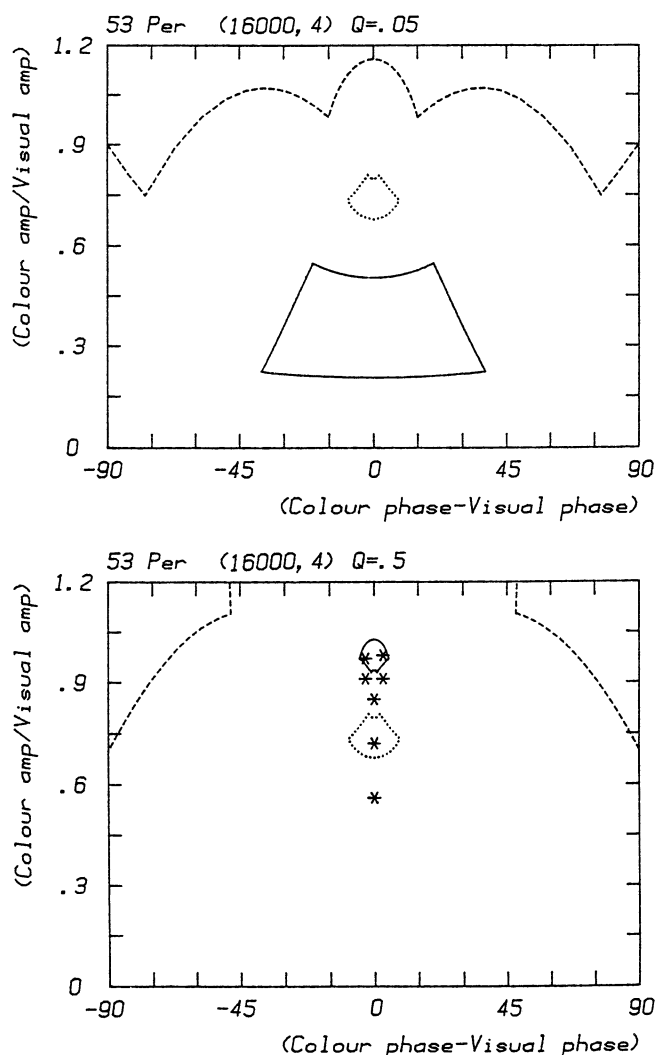


Figure 3—Similar to Figure 2 except that both model and data use Geneva photometry colours. Thus, the plot is A_{U-V}/A_V versus $(\phi_{U-V} - \phi_V)$.

give a $(A_{U-V}/A_V, \phi_{U-V} - \phi_V)$ model plot. Observed data were then assumed transformed from the Geneva system by $A_{U-V}/A_V \approx A_{U-B}/A_V$. This approximate transformation derives from consideration of the temperature dependence of the colours in the Johnson system and in the Geneva system (North and Hauk 1979). It suffers from the uncertainty that the observed flux changes in the stars may have a different wavelength dependence than that expected from temperature changes only. In Figure 3, the model plot is on the Geneva system, $(A_{U-V}/A_V, \phi_{U-V} - \phi_V)$. This was done by assuming $\alpha_T([V]) \approx \alpha_T(V)$ and $\alpha_g([V]) \approx \alpha_g(V)$, and then deriving $\alpha_T([U])$ and $\alpha_g([U])$ from the published colours of North and Hauk (1979). Observed data were then transformed on the basis that $A_{U-V}/A_V \approx 1.2 A_{U-B}/A_V$, using the temperature dependence of the colours in the Geneva system. It therefore suffers the same deficiency as the Figure 2 data transformation. The data in both Figures 2 and 3 are plotted at a nominal value of zero on the

phase difference axis. This is because the Waelkens and Rufener observations show that light and colour variations are 'in phase' but do not provide specific phase difference information.

The significant fact which emerges from both Figures 2 and 3 is that the data lie in or near the dipole and quadrupole areas of interest, which were based on the same (R, ψ_T) values as used in Figure 1. Thus, the fact that the observed colour amplitudes in the mid-B star variables are larger than those seen in the β Cep variables is not a difficulty with Waelkens and Rufener's hypothesis of non-radial pulsation in these systems. The reason for the increase in quadrupole colour amplitudes relative to dipole ones, as Q increases from 0.05 to 0.5, is simply that C has changed sign in the process. This takes the temperature and geometry terms in the light change expression from being in phase at small Q to being out of phase at large Q . At the same time, in the radial case, it is the decreasing C value with increasing Q which results in a more important geometric term compared to the temperature term in the light change expression. Because the temperature and geometry terms are always out of phase in the radial case, this decreases the denominator in A_{U-V}/A_V and so forces up the region of permitted $l=0$ values on the $(\phi_{U-V} - \phi_V)$ axis.

In conclusion, the non-radial models here appear to strengthen Waelkens and Rufener's hypothesis concerning non-radial pulsation in the mid-B star variables. However, a note of caution must be added as a result of the recent work of Balona and Laing (1986) on HR 3562. This variable was originally classed as being of the 53 Per type by Burki (1983), with colour to light amplitude ratios comparable to Waelkens and Rufener's stars. Balona and Laing demonstrated that it is, in fact, an ellipsoidal variable. Thus, other hypotheses may also be compatible with the observed colour to light amplitude ratios in the mid-B star variables.

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